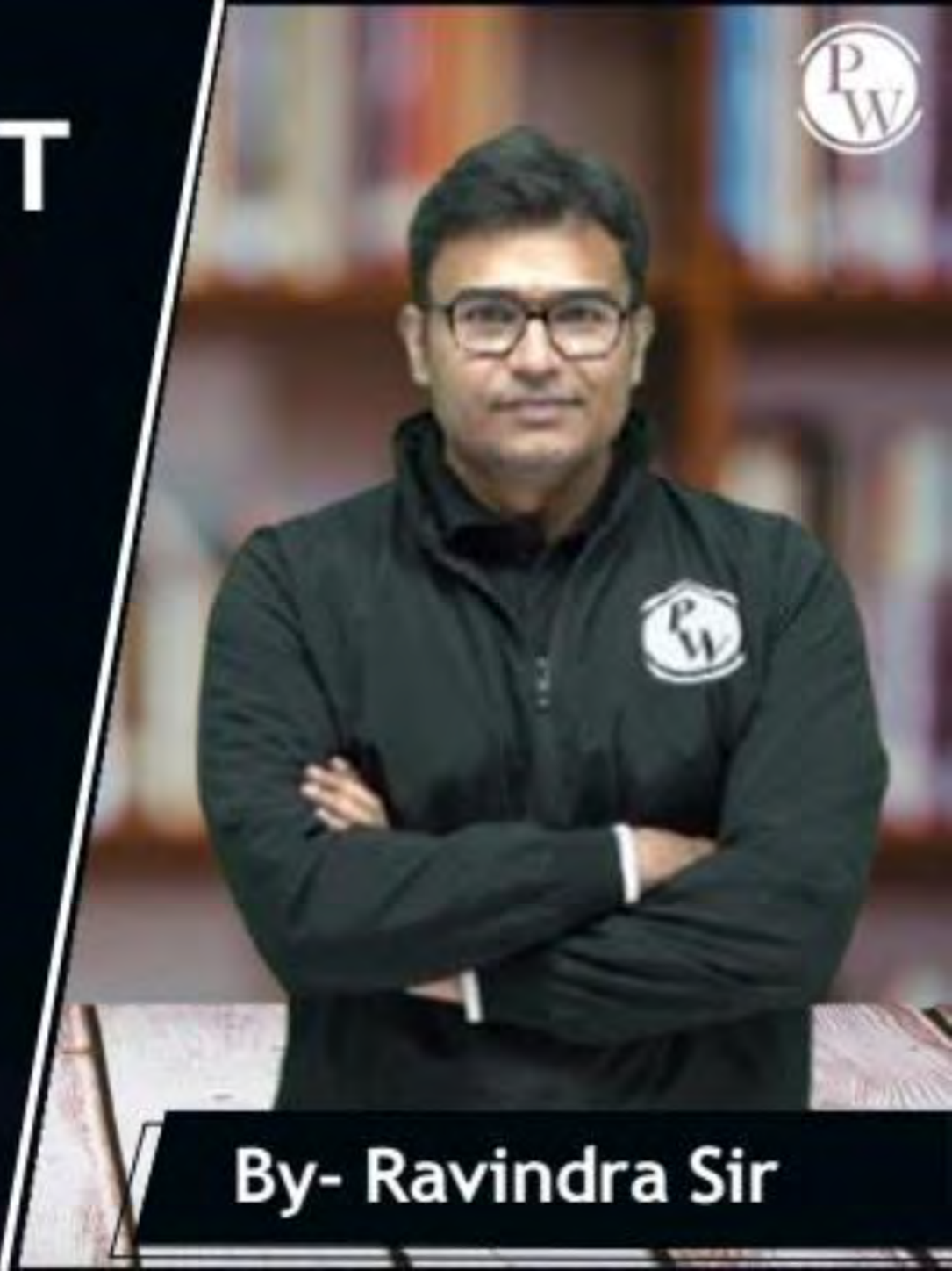
Computer Science & IT

COMPUTER NETWORK
(CN)

PYQ - CN

Lecture No. 03



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

PYQ

Topics to be Covered



- Topic
- Topic
- Topic

PYQ

A 2 km long broad cast LAN has 10^7 bps BW and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. what is the min packet size that can be used on this n/w?

a) 50 B b) 100 B c) 200 B d) None

$d = 2000m$ $BW = 10^7$ bps $v = 2 \times 10^8$ m/s.

$T_t \geq 2 * T_p$

$\frac{L}{B} \geq 2 * \frac{d}{v} \Rightarrow L \geq 2 * \frac{d}{v} * B \Rightarrow L \geq 2 \times \frac{2 \times 10^3}{2 \times 10^8} \times 10^7$

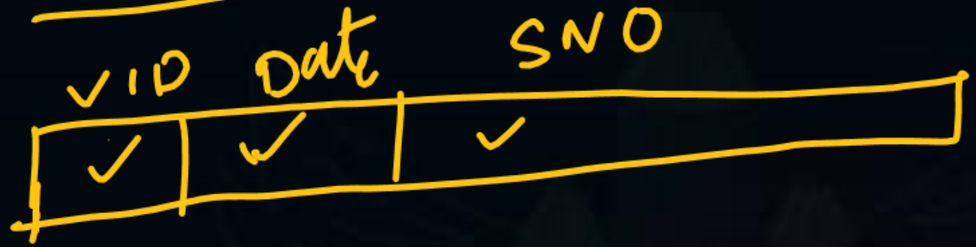
$= 200$ bits

$= 25$ B

A host is connected to a department N/W which is a part of university N/W. The university N/W in turn, is part of the Internet. The largest N/W in which ethernet address of host is unique is

a) The Subnet to which host belongs
b) The department N/W
c) The university N/W
d) the Internet

MAC



Which of the following is false regarding a bridge

a) Bridge is a layer 2 device

b) Bridge reduces collision domain

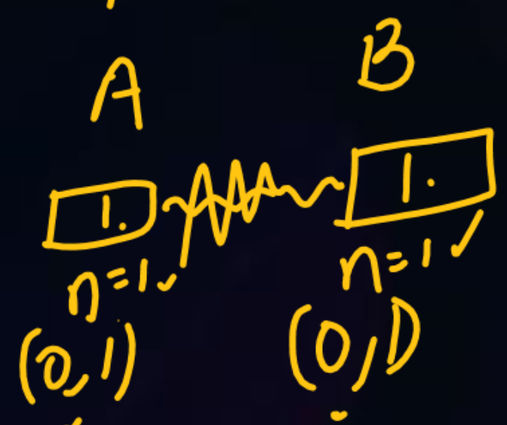
c) Bridge is used to connect two different LAN segments

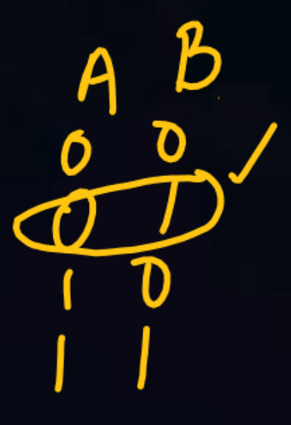
d) Bridge reduces broadcast domain.

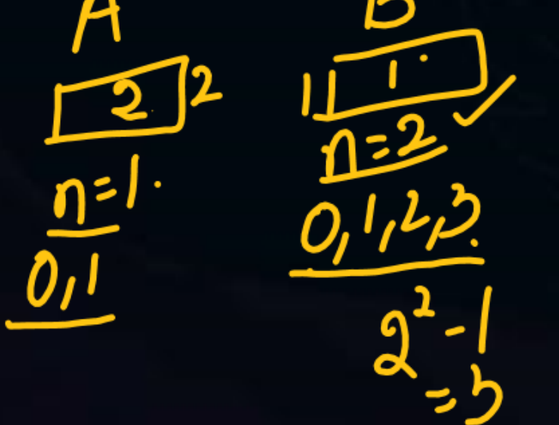


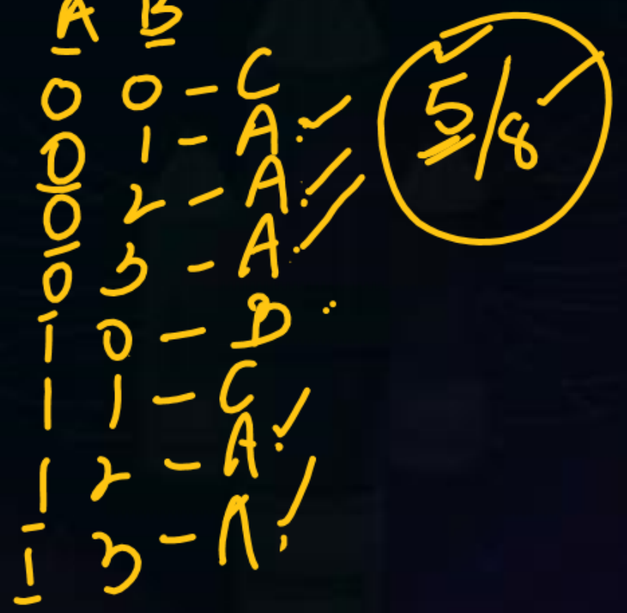
A and B are the only two stations on a Ethernet Each has a steady queue of frames. Both A and B attempt to transmit a frame, collide and A wins first backoff race. At the end of this Succesful transmission by A both A and B attempt to transmit and collide, The probability that 'A' wins the second back off race is

a) 0.5 b) 0.625 c) 0.75 d) 1.0









In Ethernet, when manchester encoding is used the bit rate is

a) Half the baud rate
b) Twice the baud rate
c) Same as the baud rate
d) None of these

$$\underline{\text{Baud}} = 2 * \underline{\text{bit}}$$

The minimum frame size required for a CSMA/CD based Computer n/w running at 1Gbps on a 200m Cable with a link speed of 2×10^8 m/s is

a) 125 B b) 250 B c) 500 B d) Non of these

$$T_t \geq 2 * T_p$$

$$\frac{L}{B} \geq 2 * \frac{d}{v}$$

$$L \geq 2 * \frac{d}{v} * B.$$

$$L \geq 2 * \frac{200m}{2 \times 10^8 m/s} * 10^9 b/s$$

$=$

THANK - YOU